

Roll No.....

MID SEMESTER EXAMINATION 2025

Name of the Program: B.Tech Civil Engineering

Semester: V

Name of the Course: Air and Noise Pollution

Course Code: TCE511

Time: 90 minutes

Max. Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub-question

(ii) Each question carries 10 marks

Q1	10 marks	COs
(a)	Discuss the following: (i) Acid Rain – source cause and its ill effects (ii) Ozone depletion	CO-1
	OR	
(b)	Classify the air pollutants based on (i) source, (ii) mode of release and (iii) chemical composition and size	CO-1
Q2	10 marks	COs
(a)	Discuss “Green House Effect”. Detailed the causes of GHE and its ill effects in the atmosphere.	CO-1
	OR	
(b)	Discuss in detail “Coal Induced Smog” that occurs naturally in atmosphere.	CO-1
Q3	10 marks	COs
(a)	In a Tabular form, mentioning Name, Description, sources, Health effects and Ill effects, provide details of the following gases: CO, PM, Pb, OZONE,	CO-1
	OR	
(b)	Discuss the major classification of Air Pollution Sources based on (i) origin, (ii) based on position	CO-1
Q4	10 marks	COs
(a)	Highlight the following Air Pollution Measurement Techniques: (i) Gravimetry (ii) Gas Chromatography (iii) Spectroscopy/ Spectrophotometry (iv) Atomic Absorption Spectroscopy (AAS) (v) Fluorometry	CO-2
	OR	
(b)	Provide definition, purpose and key features of Ambient Air Quality Standards as well as Emission Standards in relation to EPA and CPWD	CO-2
Q5	10 marks	COs
(a)	What do you understand by AQI? List the gases and particulate matters involved in calculating AQI. How AQI scale is defined?	CO-2
	OR	
(b)	Highlights the typical features of active and passive air samplers used to collect air samples for laboratory analysis.	CO-2

